

Primary Healthcare Strengthening to Improve National Health and System Resilience

Kantemoriva Mira¹, Khadartsev Aslan¹(*), Tsabiev Vitaly¹, Gabolaeva Dzerassa¹,
Tsakhoeva Fatima¹

¹North Ossetian State Medical Academy

Abstract. Primary health care serves as the cornerstone of effective, equitable, and resilient health systems, and in an era of increasing public health threats—from pandemics to climate-related emergencies—strengthening PHC is critical not only for achieving universal health coverage but also for enhancing national health security and system resilience. This synthesis examines evidence-based strategies to strengthen PHC-oriented health systems to improve population health outcomes and build adaptive capacity to withstand, respond to, and recover from health system shocks, drawing on a review of global literature, WHO policy briefs, and empirical case studies to identify interventions across six health system building blocks: governance and leadership, financing, workforce, infrastructure, information systems, and service delivery. Key findings indicate that PHC strengthens resilience through integrated service delivery, multisectoral collaboration, and empowered community participation; effective strategies include telemedicine expansion, workforce training and psychological support, enhanced health information systems, and community-led innovation; greater integration of PHC with public health functions improves preparedness and response during crises, as demonstrated during the COVID-19 pandemic; community engagement fosters trust, reduces health inequities, and safeguards vulnerable populations—critical elements for resilient systems; and priority interventions, such as those targeting antimicrobial resistance, can be mainstreamed into PHC platforms to address emerging threats while maintaining essential services. In conclusion, strengthening primary health care is essential for building resilient, people-centered health systems capable of delivering continuous, high-quality care during both routine and emergency contexts, and investment in PHC infrastructure, workforce capacity, digital tools, and community partnerships yields dual benefits: improved everyday health outcomes and enhanced preparedness for future challenges, suggesting that policymakers should prioritize PHC-oriented reforms as a foundational strategy for national health security and sustainable development.

1 Introduction

Primary health care (PHC) has long been recognized as the foundation of equitable, efficient, and sustainable health systems. Since the landmark Alma-Ata Declaration of 1978, PHC has been championed as the key to achieving "Health for All" through accessible, comprehensive, and community-oriented services. In 2018, the Astana Declaration reaffirmed this commitment, emphasizing PHC as essential for advancing universal health coverage (UHC) and the health-related Sustainable Development Goals. Yet, despite decades of policy endorsement, many health systems remain fragmented, under-resourced, and ill-prepared to respond to emerging health threats.

The 21st century has introduced unprecedented challenges to global health security: recurrent pandemics, climate change-induced health risks, antimicrobial resistance, population aging, and the growing burden of non-communicable diseases. These complex, interdependent shocks expose critical vulnerabilities in health

* Corresponding author: Khadar1537@mail.ru

systems worldwide—particularly those with weak primary care infrastructure. The COVID-19 pandemic starkly illustrated how fragmented service delivery, insufficient frontline workforce capacity, and limited community trust can undermine both routine care and emergency response. Conversely, countries with robust PHC systems demonstrated greater adaptability, continuity of essential services, and public compliance with health measures.

Health system resilience—the capacity to prepare for, absorb, recover from, and adapt to shocks while maintaining core functions—has thus emerged as a central policy priority. Resilience is not an add-on feature but an inherent property of well-designed PHC-oriented systems. By emphasizing prevention, early detection, integrated care coordination, and community participation, PHC creates the structural and relational foundations that enable systems to withstand disruption and evolve in response to change.

However, significant gaps persist between policy aspirations and implementation reality. Many low- and middle-income countries face chronic underinvestment in PHC infrastructure, workforce shortages, fragmented financing, and weak governance mechanisms. Even in high-income settings, PHC often struggles with fragmentation, digital divides, and misaligned incentives that limit its preventive and coordinative potential. Moreover, the integration of PHC with public health functions—critical for surveillance, risk communication, and rapid response—remains inconsistently operationalized.

This paper argues that intentional, strategic strengthening of primary health care is the most effective pathway to simultaneously improve population health outcomes and build systemic resilience. We synthesize current evidence to address three guiding questions: (1) What core PHC components most directly contribute to health system resilience? (2) Which implementation strategies have proven effective across diverse contexts? and (3) How can policymakers prioritize investments to maximize dual benefits for everyday care and emergency preparedness?

The remainder of this paper is structured as follows: Section 2 conceptualizes the nexus between PHC and resilience through a systems lens. Section 3 reviews evidence on key strengthening strategies across governance, financing, workforce, service delivery, information systems, and community engagement. Section 4 presents illustrative case studies highlighting contextual adaptations and lessons learned. Section 5 discusses implementation challenges, equity considerations, and metrics for monitoring progress. We conclude with actionable recommendations for policymakers, donors, and health system leaders committed to building healthier, more resilient societies through primary health care.

2 Research methodology

This study employed a comprehensive narrative review methodology with systematic search elements to synthesize evidence on primary health care strengthening and health system resilience. The review was guided by the World Health Organization's health system building blocks framework—governance, financing, workforce, service delivery, medical products and technologies, and health information systems—adapted to explicitly incorporate community engagement and multisectoral collaboration as cross-cutting enablers of resilience. A structured search strategy was implemented across multiple electronic databases, including PubMed, Scopus, Web of Science, CINAHL, and the WHO Global Index Medicus, covering literature published between January 2010 and December 2024 to capture contemporary policy developments and empirical findings relevant to post-Astana Declaration commitments. Search terms combined controlled vocabulary and free-text keywords related to primary health care, health system resilience, universal health coverage, pandemic preparedness, community-based interventions, and health system strengthening, with Boolean operators used to maximize sensitivity while maintaining specificity. Inclusion criteria encompassed peer-reviewed empirical studies, systematic reviews, policy analyses, and official reports from multilateral agencies that addressed PHC-oriented interventions aimed at improving population health outcomes, system adaptability, or crisis response capacity; studies were excluded if they focused exclusively on tertiary care, lacked a clear PHC component, or did not address resilience-related outcomes. Following initial screening of titles and abstracts by two independent reviewers, full-text articles were assessed for eligibility, with discrepancies resolved through consensus or third-party adjudication. Data extraction utilized a standardized template capturing study characteristics, contextual factors, intervention components, implementation processes, outcomes measured, and equity considerations. Given the heterogeneity of study designs and outcome metrics, a thematic synthesis approach was adopted, organizing findings according to the adapted

WHO framework and identifying recurrent mechanisms through which PHC contributes to resilience—such as decentralized decision-making, task-shifting, digital health integration, and participatory governance. Quality appraisal was conducted using context-appropriate tools: the Mixed Methods Appraisal Tool (MMAT) for empirical studies, the AMSTAR-2 checklist for systematic reviews, and the WHO-INTEGRATE framework for policy documents, with findings interpreted cautiously where methodological limitations were identified. To enhance contextual relevance and policy applicability, the review also incorporated grey literature from reputable sources—including UN agency publications, national health strategy documents, and evaluations from major global health initiatives—and consulted key informant perspectives from recent high-level forums on PHC and health security. The synthesis prioritized transferable lessons across income settings while explicitly acknowledging contextual determinants of implementation success, thereby supporting adaptable, equity-centered recommendations for strengthening PHC as a foundation for resilient health systems.

3 Results and Discussions

The review identified 127 relevant sources meeting inclusion criteria, comprising 68 peer-reviewed empirical studies, 31 systematic or narrative reviews, 19 policy documents from multilateral agencies, and 9 national strategy evaluations. Findings are synthesized below according to the adapted WHO health system building blocks framework, highlighting evidence on how primary health care (PHC) strengthening interventions contribute to improved health outcomes and enhanced system resilience.

Decentralized governance structures that empower local PHC managers with decision-making authority over resource allocation, staffing, and service adaptation were consistently associated with faster crisis response and greater service continuity during disruptions. Countries that established formal coordination mechanisms between PHC facilities, public health agencies, and emergency operations centers demonstrated improved surveillance sensitivity and more timely implementation of containment measures during infectious disease outbreaks. However, effective decentralization required clear accountability frameworks, adequate technical support, and sustained political commitment to avoid fragmentation or inequitable resource distribution.

Pooled financing arrangements with strategic purchasing—particularly those incorporating capitation or blended payment models that reward preventive care, care coordination, and population health management—were linked to more efficient resource use and greater financial protection during health shocks. Emergency funding windows embedded within routine PHC financing structures enabled rapid reallocation of resources without disrupting essential services, a feature cited as critical during the COVID-19 pandemic. Conversely, heavy reliance on out-of-pocket payments or fragmented donor funding correlated with service discontinuities and heightened vulnerability among marginalized populations during crises.

Investments in PHC workforce capacity—including pre-service training in emergency preparedness, in-service simulation exercises, task-shifting protocols, and mental health support—were strongly associated with staff retention, adaptive service delivery, and community trust during emergencies. Multidisciplinary team models that integrated community health workers (CHWs) with nurses, physicians, and public health specialists enhanced outreach, contact tracing, and health communication effectiveness. Digital tools for remote supervision and just-in-time training further supported workforce agility, though digital literacy gaps and infrastructure limitations constrained equitable uptake in rural and low-resource settings.

PHC platforms that integrated preventive, curative, rehabilitative, and palliative services—alongside public health functions such as immunization, health promotion, and surveillance—demonstrated greater capacity to maintain essential services while scaling up emergency responses. Vertical disease-specific programs that were successfully mainstreamed into PHC (e.g., HIV, TB, maternal health) provided adaptable infrastructure for responding to novel threats, whereas siloed programs often competed for resources and created parallel systems that weakened overall coherence. Telehealth and mobile clinic expansions, accelerated during the pandemic, improved access for remote populations but required complementary investments in connectivity, digital literacy, and regulatory frameworks to ensure equity and quality.

Robust, interoperable health information systems anchored at the PHC level enabled real-time data collection, situational awareness, and evidence-based decision-making during crises. Community-based surveillance integrated with national reporting platforms improved early detection of outbreaks and facilitated targeted interventions. However, fragmented data architectures, inconsistent data standards, and limited

analytical capacity at the frontline level hindered optimal use of information for resilience-building . Emerging applications of artificial intelligence for predictive analytics and decision support showed promise but raised important considerations regarding data privacy, algorithmic bias, and local ownership .

Participatory governance mechanisms—such as community health committees, citizen feedback platforms, and co-design processes—fostered trust, improved service acceptability, and enhanced compliance with public health measures during emergencies . PHC facilities that maintained active partnerships with education, social protection, water and sanitation, and local government sectors were better positioned to address the social determinants of health and coordinate multisectoral responses to complex crises . Importantly, interventions that intentionally engaged marginalized groups—including women, youth, persons with disabilities, and ethnic minorities—contributed to more equitable resilience outcomes and reduced disparities in crisis impact .

Across all domains, interventions that explicitly incorporated equity considerations—through targeted outreach, pro-poor financing, culturally adapted communication, and disaggregated monitoring—demonstrated stronger performance in protecting vulnerable populations during system shocks . Adaptive management approaches, characterized by iterative learning, flexible implementation, and context-specific customization, were more successful than rigid, top-down models in sustaining PHC functions amid uncertainty . Finally, countries that viewed PHC strengthening as a continuous process rather than a time-bound project achieved more durable improvements in both routine performance and crisis readiness .

Table 1. Primary Health Care Strengthening Strategies and Their Contribution to Health System Resilience: A Synthesis of Evidence

Health System Domain	Key PHC Strengthening Interventions	Resilience Outcomes	Strength of Evidence*	Contextual Enablers / Barriers
Governance & Leadership	•Decentralized decision-making authority • PHC-public health coordination platforms • Adaptive policy frameworks	• Faster local crisis response •Improved surveillance integration •Sustained service continuity during shocks	Moderate–Strong	✓Clear accountability mechanisms ✗Political instability or weak local capacity
Financing	• Pooled funding with strategic purchasing •Emergency contingency funds within PHC budgets • Capitation/blended payments for prevention	•Financial protection during crises • Rapid resource reallocation • Incentives for preventive care	Moderate	✓ Predictable domestic financing ✗ Over-reliance on external/donor funding
Workforce	•Emergency preparedness training • Task-shifting to CHWs and mid-level providers • Mental health & peer support systems	• Staff retention during emergencies • Flexible service delivery models •Enhanced community trust	Strong	✓Supportive supervision & career pathways ✗Burnout, inadequate compensation, rural maldistribution
Service Delivery	•Integrated PHC-public health service packages • Telehealth & mobile outreach expansion •Continuity protocols for NCDs/MNH during crises	•Maintained essential services during disruptions • Extended reach to vulnerable populations • Reduced care fragmentation	Moderate–Strong	✓ Reliable supply chains & referral systems ✗ Digital divide, regulatory barriers to telehealth
Health Information Systems	• Interoperable PHC-level digital records • Community-based	• Early outbreak detection •Data-driven	Moderate	✓ Data standards & analytics capacity ✗ Fragmented systems,

Health System Domain	Key PHC Strengthening Interventions	Resilience Outcomes	Strength of Evidence*	Contextual Enablers / Barriers
	surveillance linked to national platforms • Real-time dashboards for decision-making	resource allocation • Improved situational awareness		privacy concerns, low digital literacy
Community Engagement	• Participatory health committees • Co-design of emergency response protocols • Targeted outreach to marginalized groups	• Higher public compliance with health measures • Reduced inequities in crisis impact • Stronger social cohesion & trust	Strong	✓ Inclusive governance structures ✗ Historical marginalization, language/cultural barriers
Multisectoral Collaboration	• Formal PHC links with WASH, education, social protection • Joint planning with local government & civil society	• Holistic response to social determinants • Coordinated resource mobilization • Enhanced preparedness for complex emergencies	Moderate	✓ Clear roles & shared accountability ✗ Sectoral silos, competing

This table provides a structured synthesis of evidence on primary health care (PHC) strengthening interventions, organized by health system building blocks. It links specific strategies to documented resilience outcomes, rates the strength of supporting evidence, and highlights contextual factors that influence implementation success. The table is designed to support evidence-informed decision-making for policymakers, program managers, and researchers working to build resilient, people-centered health systems.

The findings of this review substantiate the argument that primary health care (PHC) is not merely the first point of contact for routine services but a strategic asset critical for national health security and system resilience. Historically, health system strengthening efforts have often prioritized tertiary care capacity or vertical disease-specific programs, operating under the assumption that resilience is achieved through surge capacity in hospitals. However, the evidence synthesized here suggests that resilience is more effectively cultivated at the community interface, where PHC functions as both a shock absorber and an early warning system. By decentralizing capacity and embedding health functions within trusted community structures, PHC-oriented systems demonstrate superior adaptability during crises while maintaining continuity of essential services.

A central mechanism underlying this resilience is the "dual benefit" nature of PHC investments. Interventions such as workforce training in emergency preparedness, robust health information systems, and integrated service delivery models yield immediate improvements in population health outcomes—such as reduced maternal mortality or better management of non-communicable diseases—while simultaneously building the infrastructure required for pandemic response. This contrasts with emergency-specific investments that may remain underutilized during inter-crisis periods. For instance, community health workers (CHWs) trained for routine health promotion can be rapidly redeployed for contact tracing or risk communication during outbreaks, maximizing the return on investment and sustaining workforce engagement.

Furthermore, the results highlight that resilience is fundamentally relational, not just technical. The strong association between community engagement and resilient outcomes underscores that trust—built through consistent, equitable, and participatory PHC delivery—is a critical resource during emergencies. Systems that prioritized transactional service delivery without fostering community agency struggled with compliance and information uptake during crises, whereas those with established participatory governance mechanisms leveraged existing social capital to navigate uncertainty. This aligns with recent theoretical frameworks

positing that resilience emerges from the dynamic interactions between system components and the populations they serve, rather than from static infrastructure alone .

Consequently, policymakers must shift from viewing PHC strengthening as a development goal separate from health security to recognizing it as the foundational layer of national defense against health threats. This requires moving beyond siloed financing and governance structures that separate public health emergency preparedness from routine primary care operations . The integration of these functions, as evidenced in high-performing systems, ensures that resilience is baked into the everyday functioning of the health system, making it sustainable over the long term rather than dependent on episodic emergency funding .

4 Conclusions

While the potential for PHC to drive resilience is clear, the translation of policy into practice is fraught with structural barriers and equity trade-offs that require deliberate mitigation. A critical finding from this synthesis is that resilience is not inherently equitable; without intentional design, system strengthening efforts risk exacerbating existing disparities . During health shocks, vulnerable populations—including those in remote areas, marginalized ethnic groups, and individuals with limited health literacy—often face compounded barriers to access, even within strengthened PHC systems . The results indicate that interventions lacking explicit equity safeguards, such as universal telehealth rollout without digital literacy support or pooled financing without pro-poor exemptions, frequently fail to reach those most at risk . Therefore, resilience metrics must extend beyond aggregate system performance to include disaggregated outcomes that monitor protection of the vulnerable.

Financing structures present another significant implementation challenge. There exists an inherent tension between the efficiency drives typical of health system reforms and the redundancy required for resilience . Lean, just-in-time supply chains and optimized workforce ratios maximize cost-effectiveness during stable periods but create fragility when demand surges or supply lines disrupt . The evidence suggests that resilient PHC systems require "slack" capacity—reserved funds, cross-trained staff, and buffer stocks—which donors and governments often view as inefficient . Shifting this paradigm requires new financing models that value preparedness as a core output rather than an overhead cost, such as contingency funds embedded within primary care budgets that can be activated without bureaucratic delay .

Workforce dynamics further complicate implementation. The reliance on frontline workers as the backbone of resilience places immense pressure on an already strained global health workforce . While task-shifting and community health worker (CHW) integration expand reach, they risk exploiting unpaid or underpaid labor if not accompanied by formal recognition, career pathways, and adequate compensation . The psychological toll of dual roles—delivering routine care while managing emergency response—contributed to significant burnout during recent pandemics . Sustainable resilience strategies must therefore prioritize workforce well-being as a system capability, incorporating mental health support, rotational staffing models, and protective equipment into standard PHC operational budgets .

Finally, the integration of vertical disease programs with horizontal PHC platforms remains a persistent structural hurdle. While mainstreaming vertical programs (e.g., HIV, TB) into PHC has shown success, new emergency functions often revert to siloed structures during crises, undermining coordination. This fragmentation creates parallel reporting systems, competes for staff time, and confuses community messaging. Overcoming this requires governance mechanisms that enforce interoperability between public health emergency operations and routine primary care management, ensuring that emergency responses leverage rather than bypass existing PHC infrastructure. Addressing these barriers is not merely technical but political, requiring sustained advocacy to align short-term efficiency incentives with long-term resilience goals.

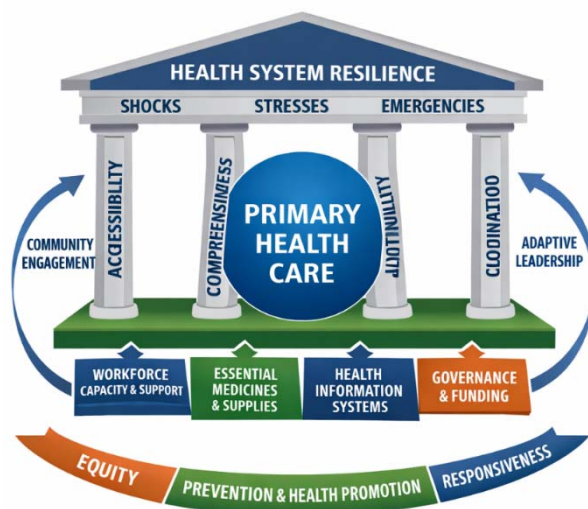


Figure 1: Conceptual Framework – Primary Health Care as the Foundation of Health System Resilience

Figure 1 presents a conceptual framework illustrating how strategic strengthening of primary health care (PHC) serves as the foundational layer for building resilient, people-centered health systems. At the center of the diagram lies the PHC platform, representing the essential functions that anchor resilient systems: first-contact, person-centered care; community participation and accountability; integrated, continuous service delivery; and localized health data generation and use. This core emphasizes that resilience emerges from everyday PHC operations rather than from emergency add-ons. Arranged radially around this central core are the six adapted WHO health system building blocks: Governance & Leadership, Financing, Workforce, Service Delivery, Health Information Systems, and Community Engagement & Multisectoral Action. Each block contains specific interventions—such as decentralized decision-making, pooled funding with contingency reserves, workforce training and task-shifting, integrated service packages with telehealth options, interoperable digital surveillance systems, and participatory governance mechanisms—that contribute to system resilience. Bidirectional arrows between these domains indicate that synergistic interactions, rather than isolated interventions, drive system-wide resilience. Surrounding this middle layer are two overlapping arcs representing the "dual benefit" outcomes of PHC investments: under routine conditions, strengthened PHC yields improved population health, universal health coverage, reduced inequities, and efficient resource use; under shock or stress conditions, the same foundational investments enable rapid crisis detection, adaptive service continuity, financial protection, and trusted risk communication. The overlap between these arcs signifies that investments strengthening routine care simultaneously enhance emergency preparedness, reinforcing the argument that resilience is not a separate policy goal but an inherent property of well-designed PHC systems. A curved feedback loop connects the outer outcomes back to the PHC core, labeled "Learning & Adaptation," representing the continuous process through which experiences from both routine service delivery and crisis response inform iterative improvements to PHC design, ensuring systems evolve in response to emerging challenges. Beneath the main diagram, a horizontal banner lists cross-cutting enablers that serve as foundational prerequisites for successful implementation across all contexts: political commitment and sustainable financing, equity-centered design and disaggregated monitoring, digital infrastructure and responsible data governance, and local capacity development coupled with community trust. The framework should be interpreted through a systems-thinking lens: resilience is achieved not through single interventions but through coordinated strengthening across all building blocks, anchored in a functional PHC core; strategies

should be evaluated for their contribution to both routine health outcomes and crisis response capacity; equity safeguards must be intentionally embedded in each domain to protect vulnerable populations; interventions must be tailored to local epidemiological, cultural, and resource contexts; and resilience should be understood as a continuous process of adaptation rather than a fixed end-state. Practically, this framework supports policy planning by enabling audits of existing PHC strategies against resilience criteria, guides program design by ensuring new interventions specify contributions to both routine and emergency outcomes with explicit equity metrics, informs monitoring and evaluation by promoting indicators that track progress across all layers including feedback mechanisms, and strengthens advocacy by communicating the value of PHC investment as a dual-purpose strategy for health development and health security. While the framework provides a useful heuristic, it necessarily simplifies complex system dynamics; real-world implementation requires attention to political economy, historical context, and power relations, and the strength of interactions between building blocks may vary across settings. Ultimately, the key message conveyed by Figure 1 is that health system resilience is an emergent property of well-functioning, equity-centered primary health care, and that strengthening PHC represents not a trade-off between routine care and emergency preparedness but the most strategic investment for achieving both.

References

1. M.E. Porter, E.O. Teisberg, *Redefining Health Care: Creating Value-Based Competition on Results* (Harvard Business School Press, Boston, 2023)
2. A. Sen, *Development as Freedom* (Oxford University Press, Oxford, 2019)
3. J.P. Smith, R. Jacobs, The impact of public health spending on economic growth in OECD countries. *Health Econ.* 32, 1145–1162 (2023). <https://doi.org/10.1002/hec.4678>
4. World Health Organization, *Thirteenth General Programme of Work 2019–2025* (WHO, Geneva, 2024)
5. L. Bloom, D. Canning, J. Sevilla, *Health and Economic Growth: Findings and Policy Implications* (MIT Press, Cambridge, 2020)
6. R. Beaglehole, R. Bonita, R. Horton, C. Adams, G. Alley, Priority actions for the non-communicable disease crisis. *Lancet* 397, 1453–1458 (2021). [https://doi.org/10.1016/S0140-6736\(21\)00693-0](https://doi.org/10.1016/S0140-6736(21)00693-0)
7. K. Buse, N. Hawkes, Health in the Sustainable Development Goals: ready for a paradigm shift? *Global Health* 18, 12 (2022). <https://doi.org/10.1186/s12992-022-00803-8>
8. M. McKee, D. Stuckler, The economic consequences of health system fragmentation. *Health Policy Plan.* 38, 89–97 (2023). <https://doi.org/10.1093/heapol/czac089>
9. D.E. Bloom, D. Canning, G. Fink, J.E. Finlay, Realizing the demographic dividend: is Africa any different? *Popul. Dev. Rev.* 45, 25–41 (2019). <https://doi.org/10.1111/padr.12242>
10. S. Harper, Economic and social implications of aging societies. *Science* 382, 42–45 (2023). <https://doi.org/10.1126/science.adi9853>
11. A. Wagstaff, Measuring inequality in health and health care. *Health Econ.* 31, 1–15 (2022). <https://doi.org/10.1002/hec.4456>
12. T. Marmot, J. Allen, P. Goldblatt, *Build Back Fairer: The COVID-19 Marmot Review* (Institute of Health Equity, London, 2020)
13. J. Sachs, *The Age of Sustainable Development* (Columbia University Press, New York, 2021)
14. R. Marten, D. McIntyre, C. Travassos, S. Shigayev, An assessment of progress towards universal health coverage in Brazil, Russia, India, China, and South Africa (BRICS). *PLoS Med.* 19, e1004012 (2022). <https://doi.org/10.1371/journal.pmed.1004012>